Ersen'S Joseph

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Summary

Mechanical Engineer specializing in moored platform design, mechanical systems, computational fluid dynamics, hydrodynamic analysis, subsea systems, hydraulic and pneumatic systems, dynamic simulation, and unmanned surface vehicle dynamics.

Technical Skills

Engineering Software: Autodesk Inventor, SolidWorks, ProteusDS, ANSYS Fluent, ABAQUS, COMSOL, MATLAB, Python

Engineering Methods: Computational Fluid Dynamics, Hydrodynamic Analysis, Dynamic Simulation, Structural Analysis, Mechanical Design, CAD, Solid Modeling, Subsea Systems Design

Manufacturing: Additive Manufacturing, Rapid Prototyping, 3D Printing, Waterjet Cutting

Professional Skills: Technical Leadership, Cross-Functional Collaboration, Project Coordination, Field Testing, Troubleshooting

Experience

Engineer II March 2026 - Present
Engineer I May 2023 - March 2026
Woods Hole Oceanographic Institution *Woods Hole, MA*

- Designed, built, and integrated mechanical systems for marine and subsea applications.
- Conducted hydrodynamic and computational fluid dynamics analyses to support engineering design decisions.
- Used CAD, numerical analysis, and simulation tools to evaluate mechanical and hydrodynamic performance.
- Supported field integration, testing, troubleshooting, and operational deployment of engineering systems.
- Collaborated with engineering teams, external vendors, and field personnel to resolve technical issues.

Production Supervisor April 2018 - May 2022
Mueller Corporation *East Bridgewater, MA*

- Led daily production operations and coordinated crews to meet production, quality, and schedule targets.
- Worked across production, quality, and maintenance teams to identify and resolve process issues.
- Supported continuous improvement activities and communicated operational priorities to production personnel.

Education

Master of Science in Mechanical Engineering Fall 2025
University of Massachusetts Dartmouth *Dartmouth, MA*

- Thesis: "Mitigation of Vortex-Induced Vibrations in Marine Stretch Hoses Using Helical Grooves"
- Advisors and Mentor: Dr. Amit Tandon, Dr. J. Thomas Farrar, and Donald B. Peters

Bachelor of Science in Mechanical Engineering May 2022
University of Massachusetts Dartmouth *Dartmouth, MA*

- Capstone Project: "Design Optimization of an Induction Coil for Drug Delivery Heating Applications"

Research and Educational Experience

Graduate Research Assistant September 2022 - September 2025
University of Massachusetts Dartmouth *Dartmouth, MA*

- Investigated vortex-induced vibrations in marine stretch hoses and evaluated helical groove designs for vibration mitigation.
- Applied hydrodynamic analysis, experimental methods, data analysis, and engineering simulation to assess system behavior.

Undergraduate Research Assistant January 2021 - May 2022
University of Massachusetts Dartmouth *Dartmouth, MA*

- Supported mechanical engineering research through experimental testing, data analysis, and technical documentation.

Teaching Assistant Fall 2022
University of Massachusetts Dartmouth *Dartmouth, MA*

Intern, Blue Economy Program Spring 2022
University of Massachusetts Dartmouth and Woods Hole Oceanographic Institution *Woods Hole, MA*

Fieldwork and Cruise Participation

Underwater Acoustic Rainfall Measurement: F/V Encourager and F/V Voyager; April 2023, October 2023, and April 2024
WFIP-3: R/V Tioga, R/V Connecticut, and M/V Captain James E. Pearce; June - December 2024
Ocean Margins Initiative: GNS Aflao, Ghana; April 2025
EKAMSAT/ASTRAL: R/V Thomas G. Thompson, Indian Ocean; May 2025
LIBS System: ROV Jason aboard R/V Atlantis, Pacific Ocean; January 2026
Ocean Margins Initiative Meteorological Tower: Ghana; July 2025 and April 2026
X-SPAR: South Africa; June 2026

Presentations

Marine Technology Society Buoy Workshop, Talks, 2024 and 2026
MUST Day, Posters, 2024 and 2025
UMass Dartmouth Mechanical Engineering Seminar, Talks, 2023 and 2025
Sigma Xi Research Exhibition, Poster, 2022
Massachusetts Undergraduate Research Conference, Poster and Talk, 2022

Certifications

Fluid Dynamics Simulation Using ANSYS, Cornell University, 2023
Responsible Conduct of Research for Engineers, CITI Program, 2020

Honors and Awards

NASA Group Achievement Award, S-MODE, 2023